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PRACTICAL JOURNAL

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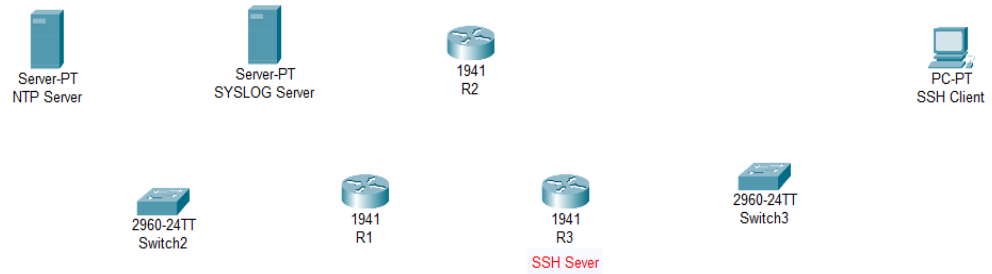
Q. Configure Routers: to log messages to the syslog server.

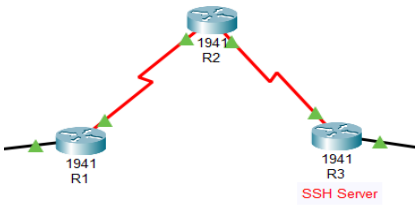
Steps:

1. Open the Cisco packet
2. Add the three routers of 1941 (R1, R2, R3) place it like triangle.
3. Click on each router (R1, R2, R3) → Go to the "Physical" tab → Turn off the router → Drag and drop the "HWIC-2T" module into an empty slot → Turn the router back on → Repeat for all routers.

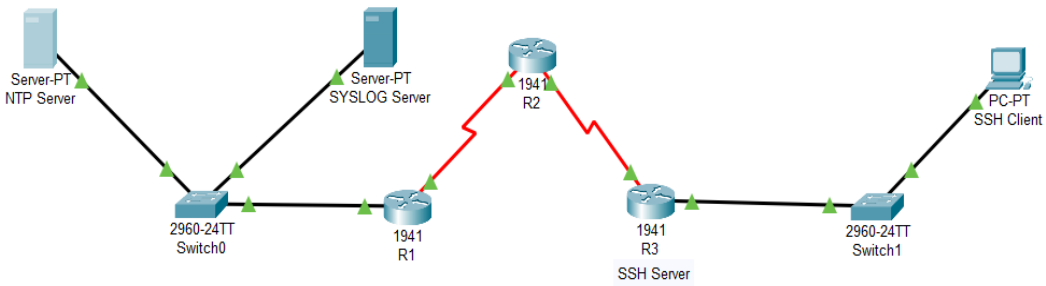
4. Take two 2960-24TT Switches → two servers → one PC and give it names as shown.



5. Connect the serial DTE wire to the router.



6. Connect the copper straight-through wire to devices as shown below.



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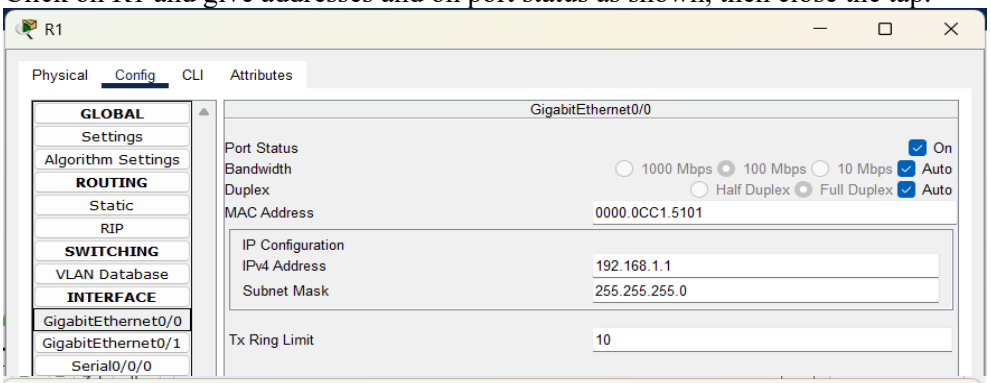
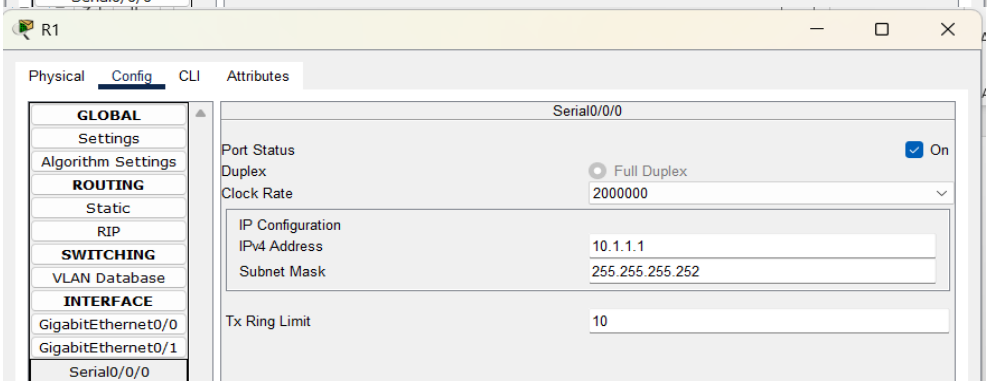
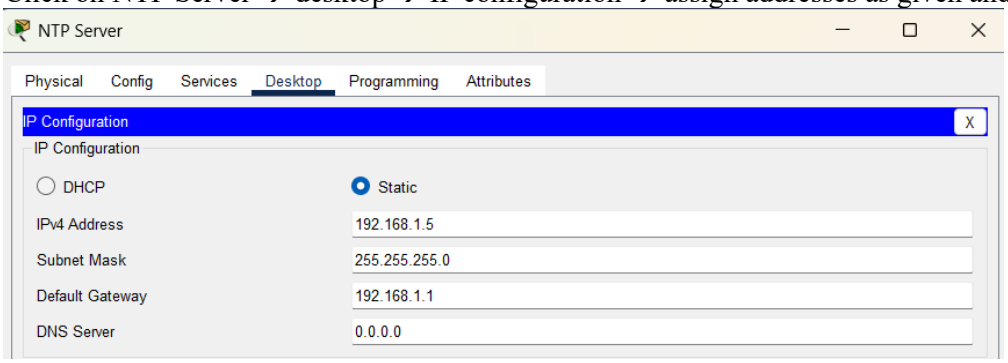
7. Click on R1 and give addresses and on port status as shown, then close the tap.

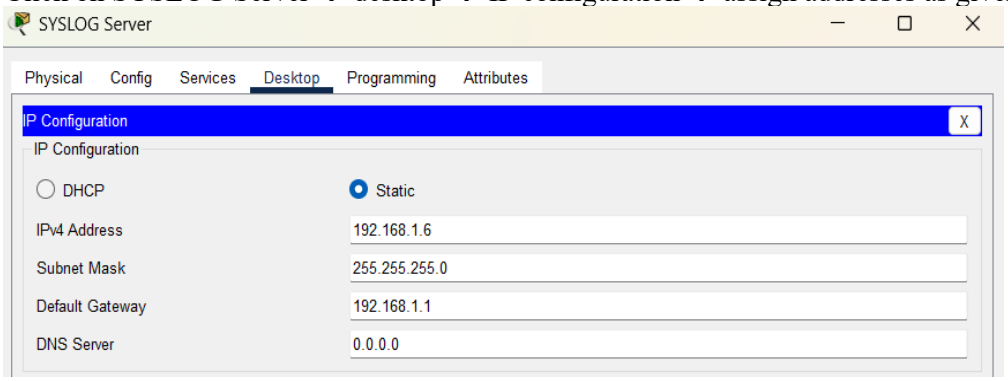




8. Click on NTP Server → desktop → IP configuration → assign addresses as given and close.



9. Click on SYSLOG Server → desktop → IP configuration → assign addresses as given and close.



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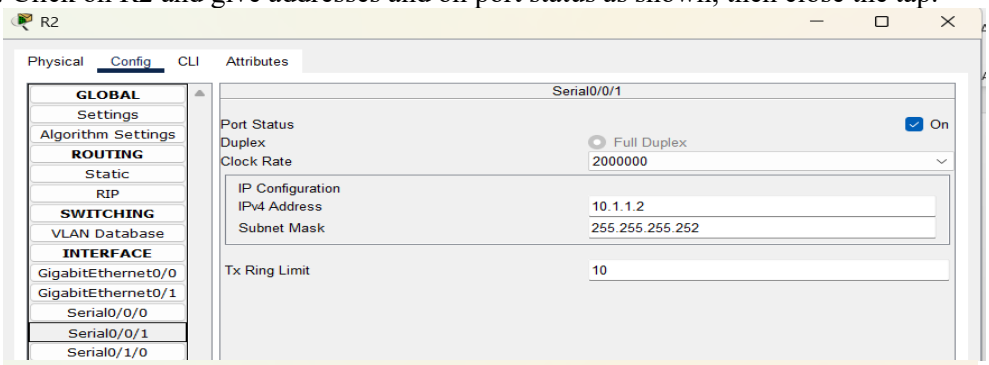
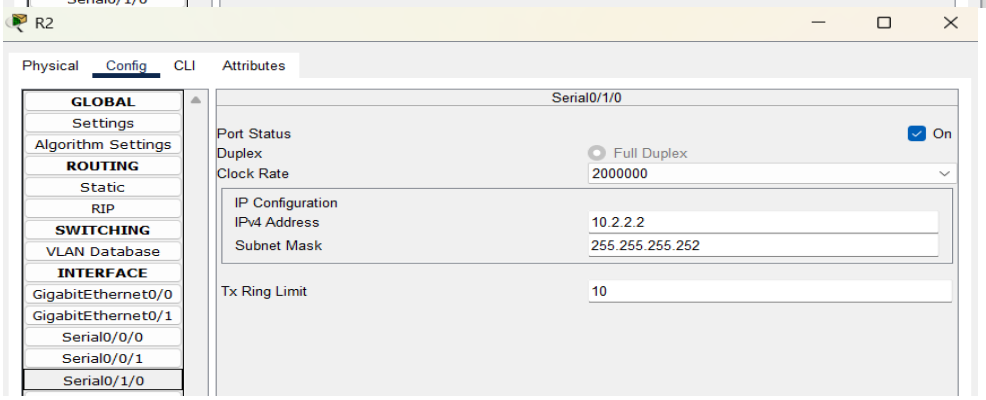
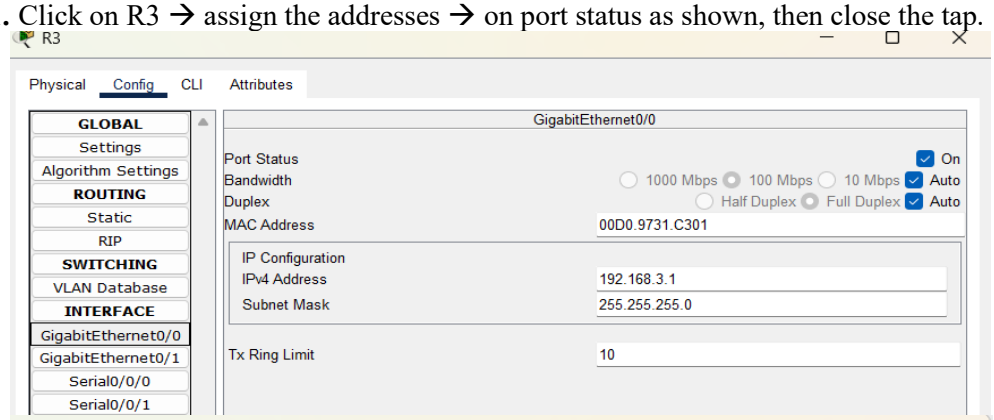
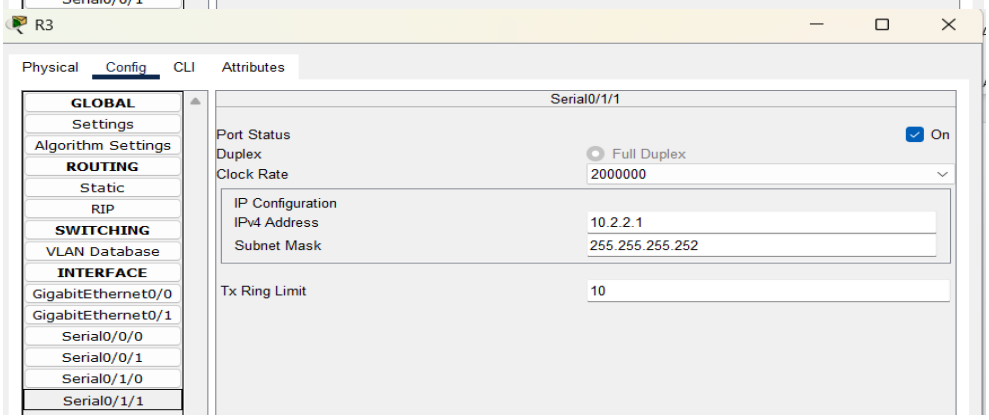
10. Click on R2 and give addresses and on port status as shown, then close the tap.





11. Click on R3 → assign the addresses → on port status as shown, then close the tap.





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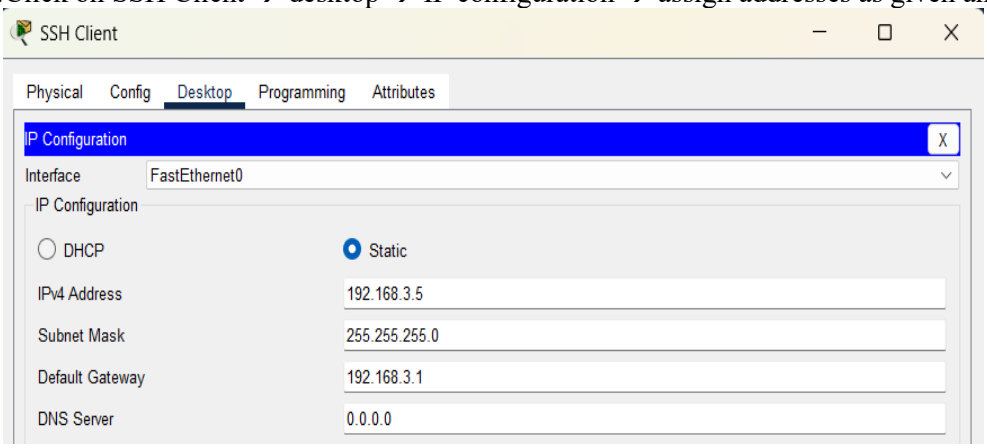
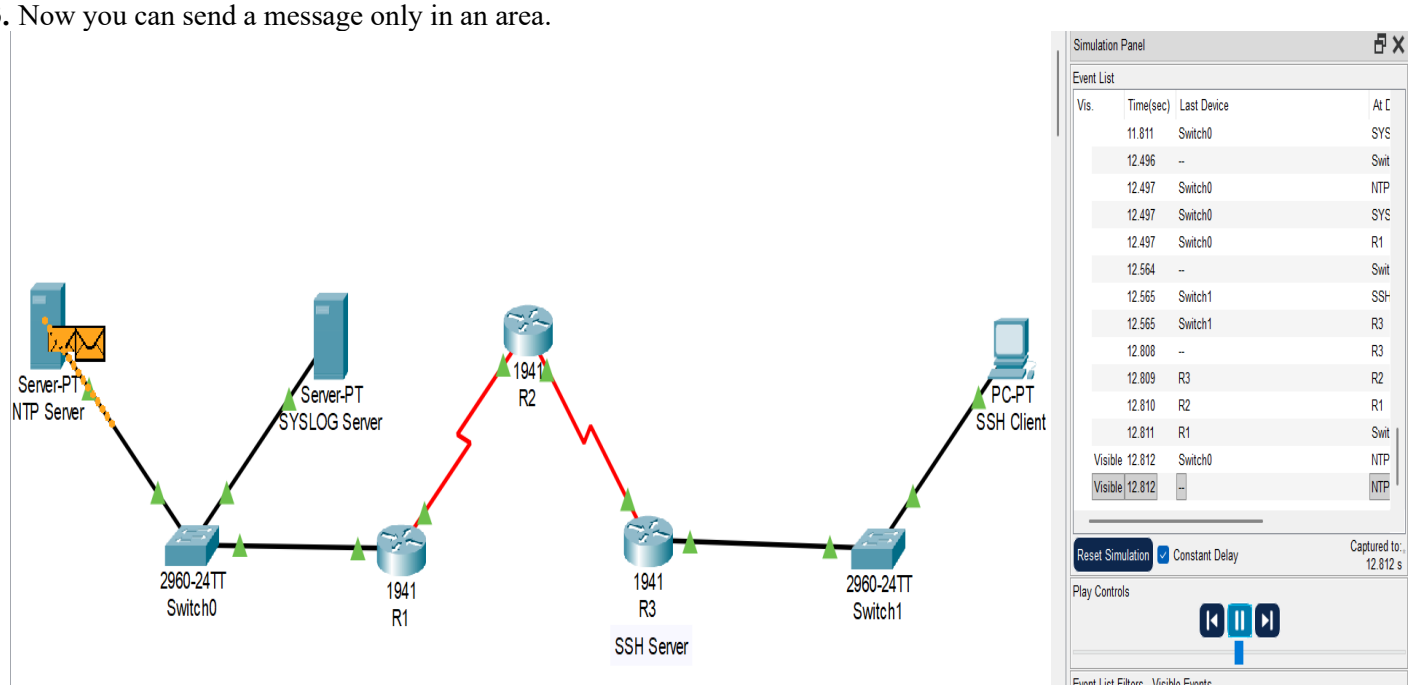
12. Click on SSH Client → desktop → IP configuration → assign addresses as given and close.



13. Now you can send a message only in an area.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	NTP ...	R1	ICMP		0.000	N	0	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	SYSL...	R1	ICMP		0.000	N	2	(edit)	(delete)
	Successful	R3	SSH Client	ICMP		0.000	N	3	(edit)	(delete)
	Successful	SSH ...	R3	ICMP		0.000	N	4	(edit)	(delete)

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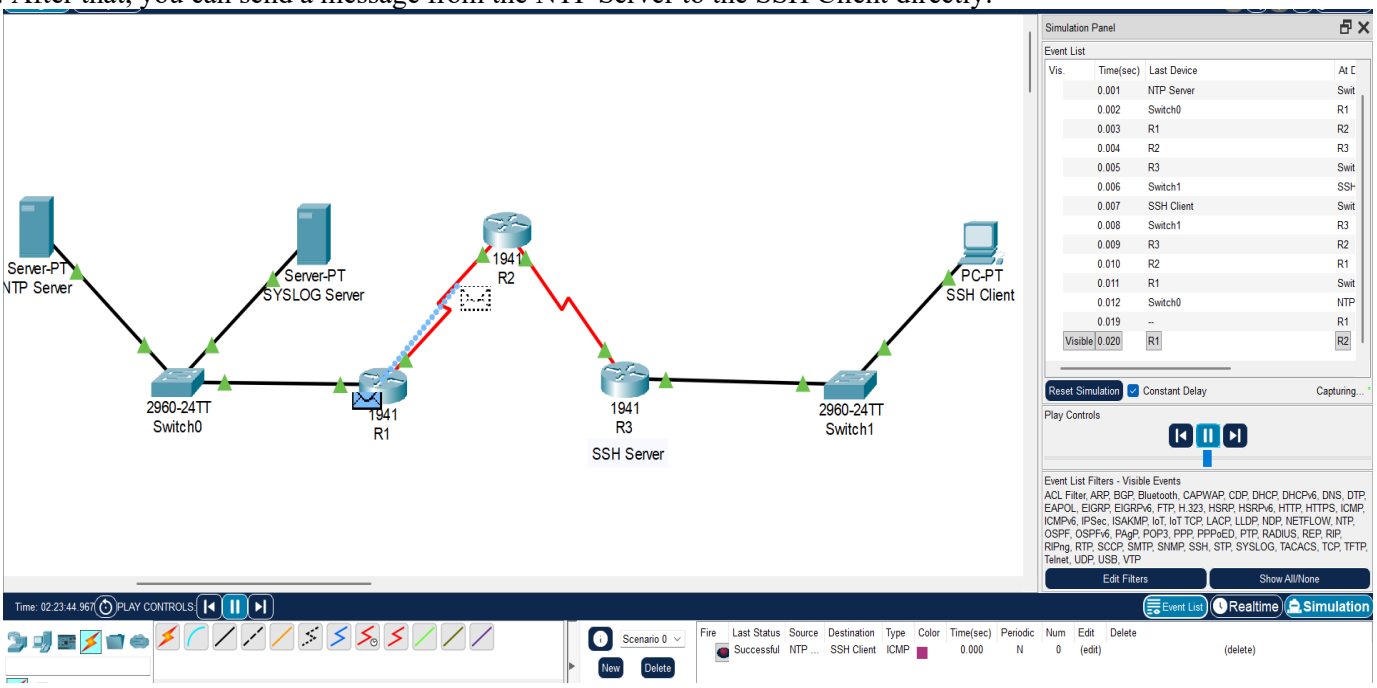
14. Click on R1 → CLI → paste these three lines in that, then close.
- router ospf 1
- network 192.168.1.0 0.0.0.255 area 0
- network 10.1.1.0 0.0.0.3 area 0

15. Click on R2 → CLI → paste these three lines in that, then close.
- router ospf 2
- network 10.1.1.0 0.0.0.3 area 0
- network 10.2.2.0 0.0.0.3 area 0

16. Click on R3 → CLI → paste these three lines in that, then close.
- router ospf 3
- network 192.168.3.0 0.0.0.255 area 0
- network 10.2.2.0 0.0.0.3 area 0

17. After that, you can send a message from the NTP Server to the SSH Client directly.



18. Ospf configuration :

R1:

```
Router(config)#line vty 0 4
Router(config-line)#password vtypa55
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret enpa55
Router(config)#
```

Ospf:

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```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#ntwork 10.1.1.0 0.0.0.3 area 0
                      ^
% Invalid input detected at '^' marker.

Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#
```

Password:

```
R1>
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#area 0 authentication message-digest
R1(config-router)#exit
R1(config)#
```

Interface:

```
R1(config)#interface s0/1/0
R1(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R1(config-if)#exit
R1(config)#
```

Show ip ospf interface:

```
R1#show ip ospf interface
  Message digest authentication enabled
    Youngest key id is 1
R1#
```

R2:

```
Router(config-if)#exit
Router(config)#line vty 0 4
Router(config-line)#password vtypa55
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret enpa55
Router(config)#
```

Ospf:

```
Router(config)#router ospf 2
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#
00:23:19: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/1/0 from LOADING to FULL,
Loading Done

Router(config-router)#network 10.2.2.0 0.0.0.3 area 0
Router(config-router)#
```

Password:

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```
R2>en
Password:
00:48:41: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/1/0 from FULL to DOWN,
Neighbor Down: Dead timer expired

00:48:41: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/1/0 from FULL to DOWN,
Neighbor Down: Interface down or detached

Password:
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#area 0 authentication message-digest
R2(config-router)#exit
R2(config)#
```

Interface:

```
R2(config)#interface s0/1/1
R2(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R2(config-if)#exit\
```

R3:

```
Router(config-if)#exit
Router(config)#line vty 0 4
Router(config-line)#password vtypa55
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret enpa55
Router(config)#
```

Ospf:

```
Router(config)#router ospf 3
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 10.2.2.0 0.0.0.3 area 0
Router(config-router)#
```

Password:

```
Router>en
Password:
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#area 0 authentication message-digest
Router(config-router)#exit
Router(config)#
```

Interface:

```
Router(config)#interface s0/1/1
Router(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
Router(config-if)#exit
Router(config)#
```

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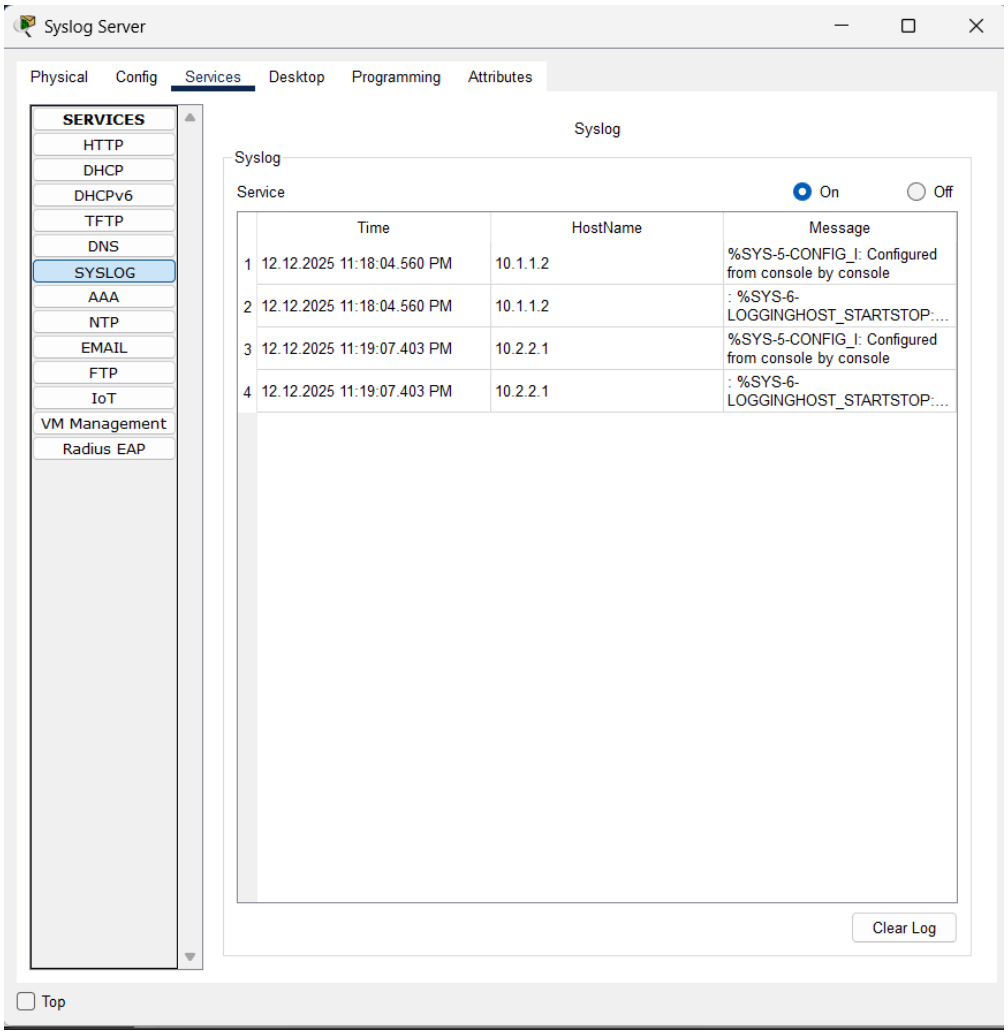
Output:

For output to be shown in Syslog server:

- Click on Syslog server → Services → SYSLOG, then turn on Service if off.
- Click on R1 → CLI → paste these lines in that, then close.
  - en
  - conf t
  - logging host 192.168.1.6
  - ex
  - show logging
- Paste these same lines on R2 and R3, then close.
- For output go to the Syslog server → Services → SYSLOG.



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